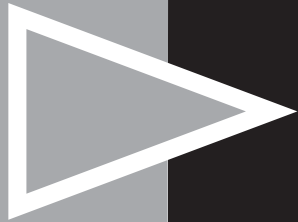




ACTIVATE BUSINESS WITH THE POWER OF I.T.™



## Care and Feeding of Your HSM Environment

Table of Contents

INTRODUCTION .....1

HSM ACTIVITIES .....2

    > Space management .....2

    > Backup.....2

    > Recycle.....2

    > Dump .....2

HSM ISSUES .....2

    > High CPU usage.....2

MAINTENANCE TASKS .....3

    > Initial setups .....3

    > Daily activities .....4

    > HSM audit .....4

BMC SOFTWARE SRM REPORTING .....5

    > Historical performance and capacity information.....5

    > Application definitions .....5

    > Space availability .....5

    > Output management.....6

    > Backup control .....6

    > Recall allocation .....6

SUMMARY ..... 7

HELPING YOU MAINTAIN ADVANTAGE ..... 7

## Introduction

Storage administrators are vital to IT departments because they protect and maintain a company's most valuable asset — its data. Storage administrators today manage increasingly larger amounts of data. That data is being stored and subsequently managed for longer periods due to lower priced storage devices and increasing governmental compliance regulations.

One of the critical roles of the storage administrator is understanding and managing DFSMSHsm, more commonly known as HSM. HSM provides backup, recovery, migration, and space management functions with automation capabilities. Storage administrators can set and apply rules within HSM to manage storage environments at both the data set level and device pool level.

Using HSM, which is shipped with z/OS, storage administrators can:

- > Ensure that space is available on DASD volumes, so that users can extend data sets and allocate new ones
- > Ensure that backup copies of data sets are available in case working copies are lost or corrupted
- > Automate manual storage management tasks
- > Improve DASD usage by managing space and data availability in a storage hierarchy
- > Provide a way to manage a data set throughout its lifecycle — from the time it is created until its last backup is no longer needed

This document outlines some best practices for working with HSM.

## HSM activities

HSM functionality includes space management (primary and secondary) and backup, recycle, and dump activity.

### Space management

Space management ensures that only *active* data resides on a direct access storage device (DASD). With increasing pressures to reduce costs, it is important to manage space efficiently and migrate unnecessary data from the primary DASD. To save space (and costs), the space management function automatically:

- > Expires data sets that have exceeded expiration dates
- > Moves unused data sets to where they can be stored less expensively, or compresses them so that they require less space
- > Frees unused portions of a data set (releases space)

Migration is the process of moving data from more-expensive volumes to less-expensive storage devices. When data is accessible to all users, it is considered level 0. Unreferenced level-0 data is migrated to a migration level-1 (ML1) disk, which is on DASD. This process is known as primary space management. The ML1 data is in a compressed format. Additional processes, such as data set compression, extent reduction, and space release, can take place during primary space management.

Secondary space management is the process of moving data from the ML1 disk to a migration level-2 (ML2) tape. While ML2 volumes can reside on DASD, HSM supports only ML2 tape. Data on tape can expire over time, depending on the expiration date of the data set or the generation of more recent backup data sets. Migrated data is recalled when a user needs it again. HSM generally processes recalled data automatically.

### Backup

The backup function backs up the HSM control data sets (CDSs), the journal, and all data sets whose change bit indicator has been flagged. These backups are generally used for local data set recovery.

This data-set-level function determines whether to copy a data set, based on the backup frequency attribute of the storage group.

### Recycle

Recycle is the process of moving valid backup or migration tape data to new tapes. When HSM expires the data residing on tape, the percentage of valid data that is residing on a specific tape decreases. The process ensures that HSM always uses full or nearly full tapes, thus maximizing tape usage.

### Dump

The dump process moves the entire contents of a disk volume to tape. Dumps are usually run weekly instead of daily and are normally sent to offsite storage for business recovery purposes.

## HSM issues

Because HSM is powerful and performs so many tasks, there is potential for problems. Common HSM problems include failures in space management, high CPU usage because of unneeded or wasteful actions, failures in the recycle process, internal control data set errors, and problems with the aggregate backup and recovery support (ABARS) or automatic dump process. Storage resource management (SRM) tools can help storage administrators discover and correct HSM related problems such as:

- > High CPU usage
- > Errors in control data sets
- > Recycle failure
- > ABARS autodump failure

### High CPU usage

High CPU usage generally raises red flags in IT shops. Using an SRM tool, you can correct some of the most common causes of high CPU usage such as:

- > Unnecessarily migrated data sets
- > Space management errors
- > Wasteful tape recycling
- > Recall thrashing

Migrating unnecessary data sets. Though many things can cause excessive CPU usage, an easy way to reduce usage is to reduce or eliminate unnecessary HSM activity created by ineffective management-class policies or application JCL. SRM tools also can help by tracking activity and thrashing, and by tying the data sets to the DFSMS constructs.

Errors in space management. You can correct errors in space management by changing management-class policies, removing undefined DSORG data sets, removing uncataloged data sets, and halting any other unnecessary HSM recall, migration, or backup activity. For example, resolving return codes 99, 19, 82, and 37 significantly increases productivity. Return code 99 is caused by an undefined DSORG, which can cause errors during backup and migration. Return code 37 is issued when there is not enough contiguous space to migrate the data set. If you receive a return code 37, you either can change the management class to prevent the large data set from going to ML1, or you can increase the amount of space available in the ML1 pool. SRM tools help you quickly see which return codes were received and how to correct them.

**Wasteful tape recycle.** Wasteful tape recycle usually happens because the “percent full” control is set too low. The recycle settings (when to run and how often) also can lead to wasteful recycles. Recycling should be part of an automated solution. It is advisable to run tape recycles during low tape-drive usage, such as early in the morning just before normal business hours, or just before the nightly batch cycle runs.

**Recall thrashing.** Recall thrashing contributes tremendously to CPU cycles. Migrating and then immediately (or shortly thereafter) recalling the same set of data sets wastes valuable HSM resources and can be quite costly. A thrashing report will help you determine what is being migrated and recalled. Examining the thrashing report, and determining who or what is recalling the data sets, will help you set better management-class policies. You also can code batch JCL to be more efficient in how it handles GDG bases or sequential files.

Small data sets are the nemesis of HSM. Even when using small data set packing (SDSP) files, the constant migration of small data sets to ML1 and then to ML2 tape — which will expire in a short period of time — is useless. Compared to the cost of DASD, file compaction in migrating is minimal. For SDSP data sets, look closely at when the data migrates and when it expires. It is a good idea to change the management-class policies for these data sets to let them “live and die” on primary disk. This reduces unnecessary migrations and the subsequent expiration from HSM. Preventing needless migration reduces the need to recycle tape volumes that have rapidly expiring data.

## Maintenance tasks

This section describes some of the tasks required to set up HSM and effectively maintain for optimum efficiency.

### Initial setups

Initial setups include creating the HSM PARMLIB, the SMS ACS routines, and the SMS constructs.

**Space management cycles.** To ensure that space management is running efficiently, determine when the primary and secondary space management cycles start and end. Do these times interfere with production batch cycles, certain online transactions, or database backup activity? Is the space management activity being completed on time? Determine the length of time it takes to run space management and adjust the space management windows to meet all business needs.

Determine if the space management cycles are spread over multiple hosts or run on the primary host. Be aware of activities that run concurrently at any given time, to ensure that they do not cause contention. By determining the location of space management, you can identify any overlap in resources. In most

cases, a single primary host is sufficient to execute all space management activities, regardless of how many LPARs are running. However, there is a limit to the number of tasks that can be executed concurrently.

Be sure to review all of the resources that are being used, including tape drives.

**Backup cycles.** You can increase efficiency of backup cycles by answering these questions:

- > Does the backup cycle, including the late-start time, interfere with any of the items mentioned in the space management discussion? If it does, adjust the times to remove the conflicts.
- > Is the backup cycle run from the primary host or multiple hosts? Your environment determines the most efficient way to run the backup cycle. Look at the number of storage groups, the average amount of data that is processed, and the number of available LPARs or HSM tasks.
- > Are the backups being performed on storage groups or pools that do not require HSM backups? Periodically examine which storage groups and pools are being backed up. There's no need to waste precious resources.

**SMS constructs.** Determine if the data sets are actually going to the proper storage groups to receive the proper HSM management. Review the SMS constructs to determine if the data has the correct management class assigned to the data sets. If the data sets have incorrect management classes, the data could be deleted before its useful lifespan ends, it could be retained for too long, or it could be in the wrong location for use (such as on an ML2 tape when the data is needed). All of these situations create unnecessary overhead, difficulty for users, and possible legal issues.

**Adding volumes and tape duplexing.** To prevent problems when adding volumes, ask these questions. Are the correct exits turned on within HSM PARMLIB? Are the proper ADDVOLS coded for non-SMS-managed volumes? If incorrect exits are being used, or if the ADDVOLS parameters are coded incorrectly, the volumes will not be added.

Consider tape duplexing for 3590-type tape cartridges, which can contain up to 80 GB of data. Trying to recover all data from a damaged ML2 tape can be time-consuming. Duplexing is an excellent solution, but be careful when determining the total number of tapes to use; using too many can fill all of the available storage capacity for the tape silo. You also will incur additional storage costs, both robotic and shelf, as well as additional tape media costs.

## Daily activities

As a storage administrator that handles many tasks, you may want to begin your daily activity by reviewing the error summary reports, activity reports, and thrashing reports.

Check the status of HSM by determining what HSM is enabled to do and what it actually is doing. A good starting point is the volume activity. Understanding how the beginning and ending thresholds look will give you a better idea of what HSM actually has processed for each volume. You can use SRM tools to drill down through the volumes to review the actual data migrated.

Check the status of the control data sets by checking HSM utilization and HSM free space.

Check the HSM work activity by looking at what completed successfully and what was unable to complete. To determine if data should not be migrated, look at the data sets that were recalled. You need to know who recalled them, when the data was recalled, and the data size. Knowing what issued the recall can help you determine if bottlenecks are being created during batch cycles. If most of the recalled data sets are small, you will probably opt to leave them on primary disk. Taking the recall one step further, the time associated with each recall will determine problems in tape libraries. If the tape comes from outside a library, or if the tape was in use for some other HSM process, the recall could be delayed.

It is important to understand what HSM is processing on a daily basis. This information can help determine where potential problems exist. Using an SRM tool and a combination of queries to HSM, you can check the following key items:

- > Space migrated
  - Volume report
  - Data set report
- > Error summary report
- > Thrashing reports
- > Recycle by type
  - 0 percent
  - 25 or 30 percent

Divide the recycle of HSM tapes into two groups. Issue the recycle for 0 percent. Then, 30 minutes later, issue the recycle for 25 or 30 percent. This process lets you reduce a larger percentage of tapes within a given time period. The 0-percent tapes do not require a tape mount.

Recycle the tapes based on the type of HSM tape only. For example, on Monday, issue a recycle only for the Monday backup cycle tapes. Do not include ML2 or any other backup cycle. This process helps reduce the number of tape drives

that are used, and it keeps the recycle process on a schedule that is easily tracked.

## HSM audit

Performing an HSM audit typically is slow and painful, but worth the effort. Auditing helps you find, and often correct, discrepancies between control data sets, catalogs, and HSM owned volumes. For example, a CDS could contain hundreds of corrected records. HSM could continue processing the corrupted CDS for days, weeks, or even months. You may not realize that a problem exists until a recycle fails or until you try to access the data set using corrupted records. By this time, the incurred damage could be costly and difficult to repair.

The audit features in HSM are difficult to use, and an audit can take days to complete. After an audit is started, you should not stop it for any reason, because the updates are not processed until after the audit has completed. All updates are nullified if the audit is stopped before completion. The best time to run an audit is once a month when other processing is slow. If you are unable to run the audit each month, run it quarterly.

CDS cleanup. The CDSs in HSM include the backup control data set (BCDS), the migration control data set (MCDS), and the offline (tape) control data set (OCDS). These CDSs have pointers to each other. The BCDS contains the backup version record (MCC), the data set record (MCB), and the eligible volume record (MCP). The MCDS contains the data set record (MCD), the alias entry record (MCA), and the volume record (MCV).

Using a good SRM tool can help you determine if there is obsolete data in the CDS. For example, you can find where HSM is backing up ISPF data sets (usually not desirable) and possibly wasting HSM resources. SRM tools can pinpoint other activity, such as the oldest data in the MCDS (from a migration standpoint), as well as which tape volumes have problems and which should be recycled manually to clean up the system.

## BMC MAINVIEW SRM Reporting

Managing HSM is not inherently easy, but SRM tools such as BMC® MAINVIEW® SRM Reporting can help. BMC MAINVIEW SRM Reporting makes it easier to work with HSM by providing:

- > Real-time HSM messages
- > Improved HSM collection database
- > Views of the active recall queue within HSM
- > Events, TSO messages, and alerts on threshold conditions including queue length, batch recalls that have waited in the queue too long, and number of recalls from the same user
- > Ability to prioritize HSM recall processing across the sysplex and the HSM Common Recall Queue (CRQ)

BMC MAINVIEW SRM Reporting provides comprehensive solutions that monitor, analyze, and automate tasks to safeguard the health of storage subsystems and ensure that critical applications complete successfully. The product provides comprehensive storage information based on real-time and interval-driven monitoring, along with commands and automated tasks. This combination of features enables storage managers to:

- > Report on storage activities as they occur
- > Manage storage subsystem performance
- > Observe trend data
- > Obtain true minimum and maximum occupancy information
- > Establish storage budgets for logical groups and warn administrators or deny allocations when budgets are exceeded
- > Obtain application and functional views of storage use
- > Quickly obtain the information most commonly used to manage disk resources
- > Define thresholds that trigger an unlimited series of self-monitoring, automatic tasks
- > Develop customized monitoring and automation procedures for volumes, pools, and logical groups

BMC MAINVIEW SRM Reporting gathers historical information on pools, volumes, data sets, and VTOCs, and displays the results. A powerful, user-defined search engine rapidly locates information that is required for daily space management.

You can use an ISPF interface or BMC® MAINVIEW® Explorer to view reports and graphs. Daily or monthly trend reports display information about:

- > Space allocated, used, and idle
- > Number of data sets
- > Fragmentation index

You can customize batch reports to select fields and the order in which they are displayed, and generate multiple reports in a single job. Virtually all information can be produced in batch.

### Historical performance and capacity information

BMC MAINVIEW SRM Reporting collects detailed historical information about storage performance and capacity. Performance statistics for data sets include response time, I/O rates, and cache activity. The product collects and summarizes performance and storage-use statistics at several levels. For example, storage managers can use monthly summaries to forecast storage usage, and use daily and interval information to pinpoint a problem. Performance reports are updated automatically at user-defined intervals (snapshots). Real-time activity information is provided for devices, I/O queuing, channels, and contention by enqueue and reserves. Vendor-specific RAID reports provide data critical to device optimization.

Performance and capacity exception management is a threshold-based facility. For example, storage managers may want to generate an alarm for net capacity load in IBM RVA disk subsystems or a high cache-read-miss percentage in EMC Symmetrix devices. When integrated with the BMC® MAINVIEW® SRM Automation product, an alarm can be sent to initiate corrective automated actions.

### Application definitions

To meet business needs, storage must be deployed and managed according to its use. Storage occupancy should be viewed from the perspective of logical groupings, such as departments or applications. BMC MAINVIEW SRM Reporting uses application definition for these user-defined groupings. DASDM exits (pre- and post-processing exits as well as user-defined exits) enable BMC MAINVIEW SRM Reporting to monitor space at allocation and deallocation to provide an accurate evaluation of an application's storage use versus its quota.

### Space availability

Controlling DASD consumption usually does not give the storage administrator authority over users. However, it may be necessary to deny an allocation for a noncritical process, so that business-critical applications have the space they need. The data sets themselves usually are not exclusive to a single job. Therefore, it is important to be able to categorize a data set in various ways and to exercise various kinds of control.

With BMC MAINVIEW SRM Reporting, a data set can be a member of as many as four application definitions that can be

tiered or hierarchical. Each application definition can have different quota controls that monitor, warn, or reject the allocation if the quota is exceeded. Important applications are assured of space, while those that are optional are restricted. You can remove restrictions when conditions change, such as when the request is made outside peak shift.

## **Output management**

You can automate the response to HSM and DFSMSdss (DSS) messages by automatically generating control statements in response to key HSM errors. Cryptic error log messages are filtered, reworded, and responded to based on your criteria. This feature saves time and makes HSM easier to manage.

## **Backup control**

BMC MAINVIEW SRM Reporting makes it easy to include or exclude data sets for backup processing. Filtering minimizes coding and ensures that automatic backups do not occur for data sets that do not need them. ALTERD commands are not required.

## **Recall allocation**

BMC MAINVIEW SRM Reporting extends the benefits of HSM to non-SMS-managed data by pooling the non-DFSMS data sets that HSM recalls. To control fragmentation within eligible pools, volumes are selected based on how the data set “best fits” the available extents.

## Summary

Periodically ask yourself the following questions:

- > Is HSM doing what I want?
- > Am I being proactive enough with HSM?
- > Does HSM have internal problems?
- > Is HSM causing outside issues?

BMC MAINVIEW SRM Reporting can help you find the answers. When you can confidently answer these questions, you have control of your HSM environment and are able to correct and manage your data efficiently and effectively.



ACTIVATE BUSINESS WITH THE POWER OF I.T.™

## About BMC Software

BMC Software helps IT organizations drive greater business value through better management of technology. Our industry-leading Business Service Management solutions ensure that everything IT does is prioritized according to business impact, so IT can proactively address business requirements to lower costs, drive revenue, and mitigate risk. BMC solutions share BMC® Atrium™ technologies to enable IT to manage across the complexity of diverse systems and processes — from mainframe to distributed, databases to applications, service to security. Founded in 1980, BMC Software has offices worldwide and fiscal 2005 revenues of more than \$1.46 billion. BMC Software. Activate your business with the power of IT. For more information, visit [www.bmc.com](http://www.bmc.com).

